STAT 540 Midterm

Name :

Problem 1:(30 points) Mark T(true) or F(false).

- (T, F) A binomial tree model is complete.
- (T, F) A conditional expectation of a random variable X_t with respect to a σ-algebra $\mathcal{G}$ is a deterministic function.
- (T, F) The martingale probability (q_u, q_d) in a binomial tree model does not depend on the original probability (p_u, p_d).
- (T, F) The Markov property tells that the future is independent of the past conditional on the current state.
- (T, F) A Brownian motion moves infinite distance in a finite time.
- (T, F) A Brownian motion has stationary and independent increments.

Problem 2:(20 points) A *binary* call option pays \$1 if the terminal stock price S_T is at or above K, and it pays nothing otherwise. Similarly, a binary put option pays \$1 if $S_T \leq K$, and pays nothing if $S_T > K$. Let $\Phi_1(S_T)$ be the payoff function of a European call option with strike price K, and let $\Phi_2(S_T)$ be the payoff function of a binary call with strike price K. Find a relation between $\Phi_2(S_T)$ and $\Phi_1(S_T)$.

Problem 3:(20 points) Explain why a one-period trinomial model is not complete.

Problem 4:(40 points) Consider a two-period binomial tree model. The initial stock price is \$100, and at each node of the tree, the stock S can either jump up by a factor $u = 1.1$ or down by a factor $d = 0.9$. Assume that interest rate $R = 0$. Consider an option whose payoff at time $T = 2$ is given by

$$2(80 - S_T)^+ + (S_T - 90)^+.$$

(1) Find the price of the option at time $t = 0$.

(2) Find values of the hedging portfolio at two nodes at time $t = 1$.

Problem 5:(20 points) Two fair coins are tossed.

(1) Write the σ-algebra generated by $\{(H, T)\}$.

(2) Let X be the number of head. Find $E(X|\{(T,H),(H,H),(T,T)\})$.

Problem 6:(20 points) Let X_j be i.i.d and $P(X_j = 1) = P(X_j = -1) = 1/2$. Also, let $M_k = \sum_{j=1}^{k} X_j$.

(1) Show that M_k is a martingale.

(2) Calculate $Var(M_k)$ and $[M, M]_k$, the quadratic variation. Discuss the difference between those two.